

Day 14 - Week 2 Peer Consolidation

Week 2 Summary

During Week 2, I mainly focused on understanding the dataset, checking its quality, performing EDA, preprocessing the data, and looking at possible feature engineering options.

Day 8 — Data Quality Check

I checked the dataset for missing values, duplicate values, and class balance.

There were no missing values in the dataset.

Many duplicate values were found, but these were kept because they are valid repeated measurements caused by the limited resolution of the ADC.

The bit column was also identified as a problem because it was the same as the target labels column. Therefore, it was removed from the input features.

The class distribution was approximately:

- Label 0: 62.5%
- Label 1: 37.5%

Day 9 — Cleaning and Data Split

The cleaning decisions were reviewed and finalized.

A stratified data split was selected:

- 70% Training data
- 15% Validation data
- 15% Test data

The stratified split was used to maintain a similar class distribution in all three datasets.

Day 10 — Exploratory Data Analysis

EDA was performed using visualizations such as histograms and boxplots.

The analysis showed that Label 0 and Label 1 are clearly separated based on the received signal value.

There was no overlap between the two classes at the currently available turbidity condition.

Based on this result, some expectations were documented for the Week 3 model experiments.

Day 11 — Preprocessing

The preprocessing steps were completed.

The dataset was divided into training, validation, and test datasets.

No additional scaling was required because the signal values were already normalized between 0 and 1.

The processed datasets were saved separately so that the same data split can be used during the upcoming model experiments.

Day 12 — Preprocessing Review

All the preprocessing decisions were reviewed.

Dataset Version 1 was finalized and will be used for the Week 3 modelling experiments.

Day 13 — Feature Engineering

Different feature engineering options were considered.

However, the current dataset has only one useful input feature, and this feature already clearly separates the two classes.

Therefore, no additional engineered features were added at this stage.

Discussion and Peer Insight

During the discussion with other interns, we compared our findings without sharing any code.

One common issue observed was that the currently available raw data is only for one turbidity or NTU condition.

However, the main objective of the project is to study the performance of the models at different turbidity levels.

Because data from higher NTU levels is not currently available, it is not yet possible to properly test how well the models perform under different turbidity conditions.

This is a limitation of the currently available dataset and not an issue caused by the preprocessing or modelling process.

Expectations for Week 3

Based on the EDA results, the two classes are already very clearly separated.

Therefore, I expect the fixed threshold method and machine learning models such as:

- Logistic Regression
- SVM
- Random Forest

to achieve very high accuracy and F1-scores on the current dataset.

If the models achieve near-perfect results, it will show that the data preprocessing and modelling process are working correctly.

However, these results alone will not show whether machine learning performs better than the fixed threshold method when turbidity increases.

To properly compare the methods under different turbidity conditions, raw signal data from higher NTU levels will be required.

This will be an important point to discuss during the next mentor review.

Week 2 Deliverables

The following work was completed during Week 2:

1. Day 8 — Data Quality and Class Audit
2. Day 9 — Cleaning and Split Review
3. Day 10 — EDA and Class Separability
4. Day 11 — Preprocessing Pipeline
5. Processed datasets:
 - processed_data/train.csv
 - processed_data/val.csv
 - processed_data/test.csv
6. Day 12 — Preprocessing Review and Dataset V1
7. Day 13 — Feature Engineering
8. Day 14 — Week 2 Peer Consolidation

The notebook UWOC_Bit_Classification_Day2.ipynb contains the EDA, data splitting, and preprocessing work completed during Week 2.